

Qualification Pack



Compressed Biogas Plant Operator

QP Code: HYC/Q4401

Version: 1.0

NSQF Level: 3.5

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Qualification Pack

Contents

HYC/Q4401: Compressed Biogas Plant Operator	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
HYC/N4401: Preparation for CBG plant operations	5
HYC/N4402: Perform the CBG plant operations	10
HYC/N9301: Working Effectively in a team	17
HYC/N9302: Maintain health, safety and security procedures	21
DGT/VSQ/N0101: Employability Skills (30 Hours)	26
Assessment Guidelines and Weightage	31
<i>Assessment Guidelines</i>	31
<i>Assessment Weightage</i>	32
Acronyms	33
Glossary	34

Qualification Pack

HYC/Q4401: Compressed Biogas Plant Operator

Brief Job Description

In the oil and gas sector, compressed biogas (CBG) operator is responsible for operating and maintaining a CBG production facility. His/her responsibilities include managing plant operations, maintaining equipment, ensuring compliance with safety regulations, keeping an eye on quality control, resolving problems, keeping records, working with the team, ensuring compliance with environmental regulations, seeking continuous improvement and taking part in training and development.

Personal Attributes

The individual should have ability to be technical expert, attention to detail, problem-solving abilities, a sense of safety, adaptability, physical fitness, teamwork, time management, environmental awareness and a dedication to lifelong learning are required of CBG plant operators.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HYC/N4401: Preparation for CBG plant operations](#)
2. [HYC/N4402: Perform the CBG plant operations](#)
3. [HYC/N9301: Working Effectively in a team](#)
4. [HYC/N9302: Maintain health, safety and security procedures](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Bio Fuel Production - Operation & Maintenance
Country	India
NSQF Level	3.5
Credits	14

Qualification Pack

Aligned to NCO/ISCO/ISIC Code	NCO2015/3134
Minimum Educational Qualification & Experience	Completed 1st year of 3-year diploma (after 10th) and pursuing regular diploma OR 8th grade pass plus 2-year NTC plus 1 Year NAC OR 11th grade pass OR 10th grade pass (with 1 years of relevant experience) OR 8th grade pass (with 3 years of relevant experience) OR Previous relevant Qualification of NSQF Level (3 with 1.5 years of relevant experience)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	31/08/2026
NSQC Approval Date	31/08/2023
Version	1.0
Reference code on NQR	QG-3.5-ES-00769-2023-V1-HSSCI
NQR Version	1.0

Remarks:

NA

Qualification Pack

HYC/N4401: Preparation for CBG plant operations

Description

This unit is about identifying different types of technical expertise in the processes and equipment, safety training, familiarity with operational procedures and equipment, knowledge of quality control procedures, familiarity with safety equipment and emergency protocols, understanding of documentation and reporting requirements as well as teamwork and communication skills are all part of preparing for CBG plant operations.

Scope

The scope covers the following :

- Identification of compressed biogas equipment
- Maintenance of compressed biogas equipment

Elements and Performance Criteria

Identification and maintenance of Compressed Biogas Equipments

To be competent, the user/individual on the job must be able to:

- PC1.** follow standard operating procedures for the manufacturing of CBG.
- PC2.** follow standard policy and procedures related CBG production.
- PC3.** check proper positioning and accessibility of Personal Protectives Equipment (PPE) at the site
- PC4.** familiarizing oneself with the plant layout and equipment.
- PC5.** prepare and maintain checklist to monitor CBG regulation by CBG plant at their work site around CBG sites.
- PC6.** identify different types of Checklists, tools, sample feeders, pumps, working of agitators, etc.
- PC7.** inspect machines, controls and equipment such as Digester heating system, mixer systems and discharge Piping Systems of a CBG plant for proper functioning.
- PC8.** fix small issues with a CBG plant's gear or equipment.
- PC9.** Identify the various substrates and its right consistency required to make up compressed biogas.
- PC10.** create and apply a thorough operational standard procedures that addresses the handling of equipments used in digestion, gas purification, compression, storage, and distribution of feedstock.
- PC11.** ensure that the organic waste used as feedstock is always available.
- PC12.** check that the anaerobic digestion should be optimised for effective biogas production.
- PC13.** schedule equipment maintenance and carry out regular inspection
- PC14.** check equipment readiness and ensuring proper maintenance before starting the operations
- PC15.** prepare and maintain daily inspection report and submit it to the reporting authority responsible for health, safety and environment of department

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** the types and applications of CBG equipment
- KU2.** components and materials used in CBG
- KU3.** safe work practices and procedures for CBG sites
- KU4.** identifying and addressing potential hazards in CBG work
- KU5.** communication skills for conveying equipment and work requirements
- KU6.** organizational procedures for safe handling of equipment and machine operations
- KU7.** reporting protocol and documentation required
- KU8.** relevant health and safety requirements applicable to the workplace
- KU9.** the importance of personal protective equipment used during the procedure performing
- KU10.** ways of efficiently managing renewable sources such as water in the process
- KU11.** latest methods of energy and material conservation
- KU12.** categorization of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics
- KU13.** the difference between recyclable/reusable and waste material

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret the standard operating procedures for Compressed Biogas work
- GS2.** Skill in working on elevated platforms, CBG plants, or Machines
- GS3.** record data on waste disposal at workplace
- GS4.** assess complexity and criticality of task
- GS5.** communicate effectively with team members
- GS6.** perform basic mathematical calculation
- GS7.** Perform basic organic matters
- GS8.** complete tasks efficiently and accurately within stipulated time
- GS9.** work constructively and collaboratively with others.
- GS10.** delegate tasks to improve efficiencies
- GS11.** plan and organize the marketing activity

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identification and maintenance of Compressed Biogas Equipments</i>	10	20	-	-
PC1. follow standard operating procedures for the manufacturing of CBG.	1	1	-	-
PC2. follow standard policy and procedures related CBG production.	1	1	-	-
PC3. check proper positioning and accessibility of Personal Protectives Equipment (PPE) at the site	1	2	-	-
PC4. familiarizing oneself with the plant layout and equipment.	1	1	-	-
PC5. prepare and maintain checklist to monitor CBG regulation by CBG plant at their work site around CBG sites.	1	2	-	-
PC6. identify different types of Checklists, tools, sample feeders, pumps, working of agitators, etc.	2	2	-	-
PC7. inspect machines, controls and equipment such as Digester heating system, mixer systems and discharge Piping Systems of a CBG plant for proper functioning.	1	3	-	-
PC8. fix small issues with a CBG plant's gear or equipment.	-	1	-	-
PC9. Identify the various substrates and its right consistency required to make up compressed biogas.	1	1	-	-
PC10. create and apply a thorough operational standard procedures that addresses the handling of equipments used in digestion, gas purification, compression, storage, and distribution of feedstock.	1	1	-	-
PC11. ensure that the organic waste used as feedstock is always available.	-	1	-	-
PC12. check that the anaerobic digestion should be optimised for effective biogas production.	-	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. schedule equipment maintenance and carry out regular inspection	-	1	-	-
PC14. check equipment readiness and ensuring proper maintenance before starting the operations	-	1	-	-
PC15. prepare and maintain daily inspection report and submit it to the reporting authority responsible for health, safety and environment of department	-	1	-	-
NOS Total	10	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4401
NOS Name	Preparation for CBG plant operations
Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Bio Fuel Production - Operation & Maintenance
NSQF Level	3.5
Credits	4
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

HYC/N4402: Perform the CBG plant operations

Description

This unit is about efficiently carrying out operations in a CBG plant. The operations involve tasks such as monitoring digester parameters, operating feeders and pumps, managing gas storage systems, conducting gas purification procedures, analyzing gas quality and ensuring compliance with safety protocols.

Scope

The scope covers the following :

- Biomass/Feedstock Storage and Handling
- Monitoring and controlling the Anaerobic Digestion process
- Maintain Bio Gas Purification and Treatment during storage
- Safety and security

Elements and Performance Criteria

Biomass/Feedstock Storage and Handling

To be competent, the user/individual on the job must be able to:

- PC1.** Coordinate with Waste storage handler to collect feedstock for the preparation of organic waste materials that serve as the input for the anaerobic digestion process
- PC2.** Sort the feedstock to remove any non-biodegradable or undesirable materials that can potentially hinder the digestion process
- PC3.** Apply pre-treatment techniques like maceration or thermal processing, grinding or shredding if needed for the sorted feedstock
- PC4.** adjust feedstock input rates to optimize gas production

Monitoring and controlling the Anaerobic Digestion process

To be competent, the user/individual on the job must be able to:

- PC5.** Operate the mechanical mixing system and start/ stop the mixers as required and adjust the speed and intensity of mixing based on the characteristics of the feedstock and the operational requirements
- PC6.** regularly monitor digester temperature, pressure, and pH levels to promote the growth and activity of anaerobic bacteria, which are responsible for the biogas production
- PC7.** optimizing digester temperature and pH for efficient digestion
- PC8.** conducting routine inspections of pumps, valves and other equipments to check and maintain proper flow rates throughout the plant system.
- PC9.** perform preventive maintenance tasks such as lubrication and filter changes.
- PC10.** report any abnormalities or equipment malfunctions to the technical and maintenance team for timely repairs.

Maintain Bio Gas Purification and Treatment during storage

To be competent, the user/individual on the job must be able to:

- PC11.** monitor and maintain desired gas purity levels by activating gas purification units to remove impurities

Qualification Pack

- PC12.** initializing compressors to increase gas pressure to reduce the volume of the biogas produced for Storage and transportation purpose
- PC13.** analyze gas composition using gas analyzers as per Standards.
- PC14.** sampling and analyzing biogas for key parameters (e.g., methane content, moisture).
- PC15.** adjusting purification settings to ensure compliance with quality standards.
- PC16.** monitor and control gas storage conditions
- PC17.** generate reports on gas production, quality, and operational performance

Safety and security

To be competent, the user/individual on the job must be able to:

- PC18.** apply emergency shutdown procedures.
- PC19.** respond promptly to alarms or abnormal conditions.
- PC20.** coordinate with the team to resolve any safety or operational issues.
- PC21.** maintaining accurate records of operational parameters (e.g., gas flow, temperature).
- PC22.** document any incidents, maintenance activities, or safety-related observations
- PC23.** report the problem to his reporting authority in case of emergency situations
- PC24.** participate in shift handovers to ensure smooth continuity of operations.
- PC25.** stay updated on new technologies and best practices in CBG plant operations.
- PC26.** seeking opportunities to improve operational efficiency and safety
- PC27.** prepare and maintain daily inspection report and submit it to the reporting authority responsible for CBG plant
- PC28.** follow standard operating procedures for reviewing safety procedures and protocols

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety protocols and their implementation in CBG plant operations
- KU2.** process flow and optimization in CBG production
- KU3.** standards requirements and checks for CBG as per SOP
- KU4.** environmental compliance and waste management in CBG plants
- KU5.** effective teamwork and communication with team
- KU6.** feedstock characteristics and anaerobic digestion process
- KU7.** potential hazards, risks and threats based on the nature of work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret the standard operating procedures for Compressed Biogas operations
- GS2.** working on elevated platforms, CBG plants or Machines
- GS3.** Interpret technical understanding of equipment operation
- GS4.** record data on waste disposal at workplace

Qualification Pack

- GS5.** assess complexity and criticality of task
- GS6.** Collaborate and communicate effectively with team members
- GS7.** perform basic mathematical calculation
- GS8.** Perform basic organic matters
- GS9.** complete tasks efficiently and accurately within stipulated time
- GS10.** work constructively and collaboratively with others.
- GS11.** plan and organize the marketing activity
- GS12.** Commitment to safety and creating a safe work environment.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Biomass/Feedstock Storage and Handling</i>	3	8	-	-
PC1. Coordinate with Waste storage handler to collect feedstock for the preparation of organic waste materials that serve as the input for the anaerobic digestion process	1	2	-	-
PC2. Sort the feedstock to remove any non-biodegradable or undesirable materials that can potentially hinder the digestion process	1	2	-	-
PC3. Apply pre-treatment techniques like maceration or thermal processing, grinding or shredding if needed for the sorted feedstock	1	2	-	-
PC4. adjust feedstock input rates to optimize gas production	-	2	-	-
<i>Monitoring and controlling the Anaerobic Digestion process</i>	7	9	-	-
PC5. Operate the mechanical mixing system and start/ stop the mixers as required and adjust the speed and intensity of mixing based on the characteristics of the feedstock and the operational requirements	1	2	-	-
PC6. regularly monitor digester temperature, pressure, and pH levels to promote the growth and activity of anaerobic bacteria, which are responsible for the biogas production	2	2	-	-
PC7. optimizing digester temperature and pH for efficient digestion	1	2	-	-
PC8. conducting routine inspections of pumps, valves and other equipments to check and maintain proper flow rates throughout the plant system.	1	1	-	-
PC9. perform preventive maintenance tasks such as lubrication and filter changes.	1	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. report any abnormalities or equipment malfunctions to the technical and maintenance team for timely repairs.	1	1	-	-
<i>Maintain Bio Gas Purification and Treatment during storage</i>	5	7	-	-
PC11. monitor and maintain desired gas purity levels by activating gas purification units to remove impurities	1	1	-	-
PC12. initializing compressors to increase gas pressure to reduce the volume of the biogas produced for Storage and transportation purpose	1	1	-	-
PC13. analyze gas composition using gas analyzers as per Standards.	1	1	-	-
PC14. sampling and analyzing biogas for key parameters (e.g., methane content, moisture).	1	1	-	-
PC15. adjusting purification settings to ensure compliance with quality standards.	1	1	-	-
PC16. monitor and control gas storage conditions	-	1	-	-
PC17. generate reports on gas production, quality, and operational performance	-	1	-	-
<i>Safety and security</i>	5	16	-	-
PC18. apply emergency shutdown procedures.	1	1	-	-
PC19. respond promptly to alarms or abnormal conditions.	-	1	-	-
PC20. coordinate with the team to resolve any safety or operational issues.	-	1	-	-
PC21. maintaining accurate records of operational parameters (e.g., gas flow, temperature).	-	1	-	-
PC22. document any incidents, maintenance activities, or safety-related observations	1	2	-	-
PC23. report the problem to his reporting authority in case of emergency situations	-	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. participate in shift handovers to ensure smooth continuity of operations.	-	1	-	-
PC25. stay updated on new technologies and best practices in CBG plant operations.	1	2	-	-
PC26. seeking opportunities to improve operational efficiency and safety	1	2	-	-
PC27. prepare and maintain daily inspection report and submit it to the reporting authority responsible for CBG plant	-	2	-	-
PC28. follow standard operating procedures for reviewing safety procedures and protocols	1	2	-	-
NOS Total	20	40	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4402
NOS Name	Perform the CBG plant operations
Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Bio Fuel Production - Operation & Maintenance
NSQF Level	3.5
Credits	5
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

HYC/N9301: Working Effectively in a team

Description

This unit is about working effectively within a team.

Scope

The scope covers the following :

- Effective team work

Elements and Performance Criteria

Effective team work

To be competent, the user/individual on the job must be able to:

- PC1.** maintain clear communication with colleagues
- PC2.** pass on information to colleagues in line with organisational requirements
- PC3.** provide support to the team members
- PC4.** respect the colleagues
- PC5.** fulfil commitments made to colleagues
- PC6.** inform team members timely, if timelines can't be met
- PC7.** take the necessary initiatives to resolve the issues while working in team
- PC8.** adopt gender neutral behaviour while interacting with colleagues
- PC9.** offer assistance to a person with disability (PWD), only if required

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization policies and procedures related to team performance
- KU2.** the importance of effective communication and establishing good working relationships with colleagues
- KU3.** the importance of creating an environment of trust and mutual respect
- KU4.** the implications of own work on the work and schedule of others
- KU5.** the standard practices in organisation w.r.t communication at various levels
- KU6.** the personal responsibility for completing the task in time
- KU7.** importance of gender equality
- KU8.** importance of showing empathy while interacting with a PwD

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** communicate effectively in writing
- GS2.** read instructions, guidelines/procedures
- GS3.** work in a disciplined manner for meeting commitments and deadline
- GS4.** how to plan and prioritise the work
- GS5.** the importance of consistent and reliable services
- GS6.** apply problem solving approaches in different situations
- GS7.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Effective team work</i>	20	30	-	-
PC1. maintain clear communication with colleagues	2	3	-	-
PC2. pass on information to colleagues in line with organisational requirements	2	3	-	-
PC3. provide support to the team members	2	4	-	-
PC4. respect the colleagues	3	4	-	-
PC5. fulfil commitments made to colleagues	2	3	-	-
PC6. inform team members timely, if timelines can't be met	2	4	-	-
PC7. take the necessary initiatives to resolve the issues while working in team	3	4	-	-
PC8. adopt gender neutral behaviour while interacting with colleagues	2	2	-	-
PC9. offer assistance to a person with disability (PWD), only if required	2	3	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9301
NOS Name	Working Effectively in a team
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4
Credits	2
Version	3.0
Last Reviewed Date	16/04/2026
Next Review Date	16/04/2029
NSQC Clearance Date	16/04/2026

Qualification Pack

HYC/N9302: Maintain health, safety and security procedures

Description

This unit is about maintaining health, safety and security procedure at workplace. It covers responsibilities towards self, others, assets and the environment.

Scope

The scope covers the following :

- Follow health and safety measures
- Follow safety procedures during emergency

Elements and Performance Criteria

Follow health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment such as face mask, hand gloves, goggle etc for specific tasks and work conditions
- PC2.** identify the people responsible for maintaining health and safety in the workplace
- PC3.** identify possible causes of risk or accident in the workplace
- PC4.** follow safe working practices while dealing with hazards to ensure the safety of self and others
- PC5.** lift heavy objects safely using correct procedures
- PC6.** follow safety signages
- PC7.** maintain hands hygiene by washing hand frequently and thoroughly with soap and water or alcohol-based hand rub
- PC8.** inform the concerned person of any illness related to self and others
- PC9.** maintain workplace hygiene by disinfecting the equipment and tools regularly

Follow safety procedures during emergency

To be competent, the user/individual on the job must be able to:

- PC10.** respond promptly and appropriately to an accident or in an emergency situation
- PC11.** use appropriate fire extinguishers for different types of fires correctly
- PC12.** follow appropriate rescue techniques during fire hazard
- PC13.** follow good housekeeping practice in order to prevent fire hazards
- PC14.** inform fire safety department about any near-miss incidents in the work place
- PC15.** provide appropriate first aid to victims in an emergency situation
- PC16.** follow the applicable regulations and codes as per safety standard
- PC17.** prepare written accident/incident report and share with the concerned officer/department

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** company's policies on personnel management and duty reporting procedure
- KU2.** reporting structure within organization
- KU3.** health and safety hazards commonly affecting the work environment and related precautions
- KU4.** importance of maintaining personal hygiene using PPE kit, sanitizer and soap
- KU5.** importance of maintaining workplace hygiene
- KU6.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU7.** importance of using protective clothing/equipment while working
- KU8.** various causes of fire
- KU9.** techniques of using different types of fire extinguishers
- KU10.** different materials used for extinguishing fire
- KU11.** various types of safety signs and their significance

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively by writing
- GS2.** read instructions, guidelines/procedures and reports
- GS3.** identify and report potential sources of danger
- GS4.** how to plan the work to meet the deadline
- GS5.** the importance of on time services
- GS6.** apply problem solving approaches in different situations
- GS7.** apply balanced judgments in different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow health and safety measures</i>	9	15	-	-
PC1. use protective clothing/equipment such as face mask, hand gloves, goggle etc for specific tasks and work conditions	1	2	-	-
PC2. identify the people responsible for maintaining health and safety in the workplace	1	-	-	-
PC3. identify possible causes of risk or accident in the workplace	1	2	-	-
PC4. follow safe working practices while dealing with hazards to ensure the safety of self and others	1	2	-	-
PC5. lift heavy objects safely using correct procedures	1	2	-	-
PC6. follow safety signages	1	2	-	-
PC7. maintain hands hygiene by washing hand frequently and thoroughly with soap and water or alcohol-based hand rub	1	2	-	-
PC8. inform the concerned person of any illness related to self and others	1	1	-	-
PC9. maintain workplace hygiene by disinfecting the equipment and tools regularly	1	2	-	-
<i>Follow safety procedures during emergency</i>	11	15	-	-
PC10. respond promptly and appropriately to an accident or in an emergency situation	1	2	-	-
PC11. use appropriate fire extinguishers for different types of fires correctly	2	2	-	-
PC12. follow appropriate rescue techniques during fire hazard	1	2	-	-
PC13. follow good housekeeping practice in order to prevent fire hazards	1	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. inform fire safety department about any near-miss incidents in the work place	2	2	-	-
PC15. provide appropriate first aid to victims in an emergency situation	1	2	-	-
PC16. follow the applicable regulations and codes as per safety standard	1	2	-	-
PC17. prepare written accident/incident report and share with the concerned officer/department	2	2	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9302
NOS Name	Maintain health, safety and security procedures
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4
Credits	2
Version	3.0
Last Reviewed Date	16/04/2026
Next Review Date	16/04/2029
NSQC Clearance Date	16/04/2026

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 50% of % aggregate marks to successfully clear the assessment.

Qualification Pack

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 50

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HYC/N4401.Preparation for CBG plant operations	10	20	-	-	30	25
HYC/N4402.Perform the CBG plant operations	20	40	-	-	60	25
HYC/N9301.Working Effectively in a team	20	30	-	-	50	20
HYC/N9302.Maintain health, safety and security procedures	20	30	-	-	50	20
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	90	150	-	-	240	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure
SS	Stainless Steel
PPE	Personal Protective Equipment

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.